

Notes: Barrot, Bonelli, Grassi & Sauvagnat (JFE, 2024)

Causal Effects of Closing Businesses in a Pandemic

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1 Big picture

- **Question.** What are the causal economic and health effects of state-mandated business closures during the first wave of Covid-19 in the United States?
- **Core policy.** States issued Executive Orders requiring non-essential businesses to close physical operations, with variation across states, sectors, and time.
- **Economic mechanism.** Business closures restrict the labor that can work on site. Firms lose sales and profits; workers lose wages, hours, and employment; the incidence can be shared between firms and workers.
- **Health mechanism.** Restricting workplace activity reduces person-to-person contacts, lowering infections, deaths, hospitalizations, and years of life lost, especially in contact-intensive places.
- **Identification.** The paper uses within-state and within-sector variation in the share of restricted labor. Firm regressions include firm fixed effects, sector-quarter fixed effects, and state-quarter fixed effects; CZ regressions include CZ fixed effects and state-week or state-month fixed effects.
- **Main firm result.** A 10 percentage point increase in restricted labor is associated with roughly a 2.3% decline in sales, a 1.2% decline in assets, and a 0.16 percentage point decline in quarterly ROA.
- **Main labor result.** A 10 percentage point increase in restricted labor is associated with about a 9.5% drop in wages, a 5.6% drop in employment, and an 8.2% drop in hours.
- **Main health result.** Closures significantly reduce Covid-19 infections, Covid-19 deaths, total deaths, years of life lost, quality-adjusted years of life lost, and hospitalizations.
- **Aggregate incidence.** Aggregate profit losses are around \$359 billion; labor income losses are around \$173 billion; roughly two thirds of the economic cost is borne by firms and one third by workers.
- **Cost-benefit message.** The SIRD-based estimates imply large health benefits that can exceed economic costs under plausible values of life and life-years, but uncertainty is large and linear specifications do not statistically reject net losses.
- **Policy takeaway.** Business closures were costly, but they also saved lives. Targeting closures toward high-contact areas could potentially preserve health benefits while reducing economic costs.

2 Data and measurement

2.1 Mandatory business-closure data

The authors manually read state Executive Orders to identify sectors whose physical operations were forced to close. They record issue dates, effective dates, initial end dates, eased dates, and amendments. Forty-five states issued business-closure orders between March 19 and April 6, 2020.

Interpretation: This data construction is central. The paper does not use a generic lockdown dummy; it constructs a state-sector-week closure panel that can be mapped to the employment structure of each firm and commuting zone.

2.2 Restricted labor at the firm and commuting-zone levels

$$\begin{aligned} \text{RestrictedLabor}_f &= \sum_{ind} \sum_{state} \omega_{ind,state}^f \text{Closed}_{ind,state} (1 - WFH_{ind}). \\ \text{RestrictedLabor}_{cz} &= \sum_{ind} \sum_{c \in cz} \omega_{ind,c}^{cz} \text{Closed}_{ind,c,state} (1 - WFH_{ind}). \end{aligned}$$

Interpretation: The treatment is an intensity measure: it rises with the share of jobs in closed sectors that cannot be performed remotely.

2.3 Outcomes and controls

Firm outcomes are sales, assets, and profits scaled by 2018 same-quarter values. CZ-week labor outcomes are log wages, log employment, and log hours. Health outcomes include Covid-19 infections, Covid-19 deaths, total deaths, years of life lost, quality-adjusted years of life lost, and hospitalizations. Controls include initial infections, urban share, population density, work-from-home share, contact-intensive share, children share, census controls, hospital capacity, ICU beds, Trump vote share, stay-at-home orders, and mask mandates.

Interpretation: The outcome set lets the paper estimate both sides of the pandemic trade-off: economic losses and health benefits.

3 Models and empirical specifications

3.1 Firm-level specification

$$Y_{f,t} = \mu + \xi RestrictedLabor_f \times I_{ShutDown,t} + \sigma_f + \delta_{industry \times t} + \tau_{state \times t} + \epsilon_{f,t}.$$

Interpretation: The coefficient ξ is identified from variation in how much of a firm's workforce is restricted, conditional on firm, sector-quarter, and state-quarter fixed effects.

3.2 CZ-level labor specification

$$Y_{cz,t} = \alpha_{cz} + \lambda_{state \times t} + \beta RestrictedLabor_{cz,t} + \Gamma Controls_{cz} \times I_{ShutDown,t} + u_{cz,t}.$$

Interpretation: The coefficient compares CZs in the same state and week that differ in exposure because of local industry composition.

3.3 SIRD-motivated health specification

$$I_{t+1} - I_t = \alpha I_t S_t - \beta I_t.$$

$$Health_{cz,t+\ell} = \alpha_{cz} + \lambda_{state \times t} + \beta RestrictedLabor_{cz,t} + \gamma Infections_{cz,t} + \Gamma Controls_{cz} \times I_{ShutDown,t} + \epsilon_{cz,t}.$$

Interpretation: The lagged infection term captures epidemic dynamics and behavioral/policy responses to disease prevalence.

3.4 Dynamic event-study specification

$$Y_{g,t} = \alpha_g + \lambda_t + \sum_k \beta_k RestrictedLabor_g \times \mathbf{1}\{event\ time = k\} + controls + \epsilon_{g,t}.$$

Interpretation: The event study checks whether outcomes move only after closure implementation and not before.

4 Identification and empirical design

The design uses variation across states, sectors, and time, combined with pre-determined firm and CZ employment composition. Firm regressions include firm fixed effects, sector-quarter fixed effects, and state-quarter fixed effects. CZ regressions include CZ fixed effects and state-week or state-month fixed effects. The identifying assumption is that, absent closures, changes in firm outcomes, labor income, and health outcomes would not be systematically correlated with restricted-labor exposure after conditioning on fixed effects and controls.

4.1 Why the design is credible

- Event studies show limited pre-trends before restrictions.
- Stay-at-home and mask mandates are controlled for directly.
- Health specifications include lagged infections to account for disease dynamics.
- Robustness tests include manufacturing firms, local demand controls, migration adjustment, neighboring-CZ spillovers, BLS employment validation, and Rambachan–Roth sensitivity bounds.

Interpretation: The strongest evidence comes from the combination of within-state/within-sector comparisons and dynamic pre-trend tests. The main threats are endogenous policy timing, local demand shocks, compliance differences, spillovers, and voluntary behavioral responses.

5 Tables and figures

5.1 Table 1: Timing of initial Executive Orders restricting business activities

Read: Table 1 lists issue, effective, end, extension, and easing dates by state.

Economic message: This table defines the policy timing variation used throughout the paper.

Detailed interpretation: The variation across states is essential because the paper is not estimating the effect of a single national shutdown; it uses staggered, state-specific closure exposure.

Table 1
Timing of initial Executive Orders restricting business activities. This table presents the issue date, effective date, and end date of the initial Executive Order of each U.S. state restricting business activities aside from restaurants and recreational facilities.

State	Issue date	Effective date	End date	Extended?	Eased date	End date
ALABAMA	April 3, 2020	April 4, 2020	April 30, 2020			
ALASKA	March 27, 2020	March 28, 2020	April 11, 2020	1	April 24, 2020	
ARIZONA	March 30, 2020	March 31, 2020	April 30, 2020	1	May 16, 2020	
ARKANSAS						
CALIFORNIA	March 19, 2020	March 19, 2020			May 8, 2020	
COLORADO	March 25, 2020	March 26, 2020	April 11, 2020	1	April 27, 2020	
CONNECTICUT	March 20, 2020	March 23, 2020	April 22, 2020	1	May 20, 2020	
DELAWARE	March 22, 2020	March 24, 2020	May 15, 2020		May 8, 2020	
DC	March 24, 2020	March 25, 2020	April 24, 2020	1		
FLORIDA	April 1, 2020	April 3, 2020	April 30, 2020	1	May 4, 2020	
GEORGIA	April 2, 2020	April 3, 2020	April 13, 2020	1	April 24, 2020	
HAWAII	March 24, 2020	March 25, 2020	April 30, 2020	1	May 7, 2020	
IDAHO	March 25, 2020	March 25, 2020	April 15, 2020	1	May 1, 2020	
ILLINOIS	March 20, 2020	March 21, 2020	April 7, 2020	1		
INDIANA	March 23, 2020	March 25, 2020	April 6, 2020	1	May 2, 2020	
IOWA						
KANSAS	March 28, 2020	March 30, 2020	April 19, 2020	1	May 4, 2020	
KENTUCKY	March 25, 2020	March 26, 2020			May 11, 2020	
LOUISIANA	March 22, 2020	March 23, 2020	April 13, 2020	1	May 15, 2020	
MAINE	March 24, 2020	March 25, 2020	April 8, 2020	1	May 1, 2020	
MARYLAND	March 23, 2020	March 23, 2020			May 7, 2020	
MASSACHUSETTS	March 23, 2020	March 24, 2020	April 7, 2020	1	May 18, 2020	
MICHIGAN	March 23, 2020	March 24, 2020	April 13, 2020	1	May 7, 2020	
MINNESOTA	March 25, 2020	March 28, 2020	April 10, 2020	1	April 27, 2020	
MISSISSIPPI	April 1, 2020	April 3, 2020	April 20, 2020	1	April 27, 2020	
MISSOURI	April 3, 2020	April 6, 2020	April 24, 2020	1	May 4, 2020	
MONTANA	March 26, 2020	March 28, 2020	April 10, 2020	1	April 27, 2020	
NEBRASKA						
NEVADA	March 20, 2020	March 21, 2020	April 16, 2020	1	May 9, 2020	
NEW HAMPSHIRE	March 26, 2020	March 28, 2020	May 4, 2020	1	May 4, 2020	
NEW JERSEY	March 21, 2020	March 21, 2020				
NEW MEXICO	March 23, 2020	March 24, 2020			May 15, 2020	
NEW YORK	March 20, 2020	March 22, 2020	April 17, 2020	1	May 15, 2020	
NORTH CAROLINA	March 27, 2020	March 30, 2020	April 29, 2020	1	May 8, 2020	
NORTH DAKOTA						
OHIO	March 22, 2020	March 24, 2020	April 6, 2020	1	May 4, 2020	
OKLAHOMA	March 24, 2020	March 26, 2020	April 30, 2020	1	April 24, 2020	
OREGON	March 23, 2020	March 24, 2020			May 15, 2020	
PENNSYLVANIA	March 19, 2020	March 23, 2020			May 8, 2020	
RHODE ISLAND	March 28, 2020	March 30, 2020	April 13, 2020	1	May 9, 2020	
SOUTH CAROLINA	March 31, 2020	April 1, 2020	April 15, 2020	1	May 18, 2020	
SOUTH DAKOTA						
TENNESSEE	March 30, 2020	April 1, 2020	April 14, 2020	1	May 8, 2020	
TEXAS	March 31, 2020	April 2, 2020	April 30, 2020		May 1, 2020	
UTAH						
VERMONT	March 24, 2020	March 25, 2020	April 15, 2020	1	May 4, 2020	
VIRGINIA	March 23, 2020	March 25, 2020			May 15, 2020	
WASHINGTON	March 23, 2020	March 25, 2020	April 8, 2020	1	May 15, 2020	
WEST VIRGINIA	March 23, 2020	March 24, 2020			May 4, 2020	
WISCONSIN	March 24, 2020	March 25, 2020	April 24, 2020	1	April 29, 2020	May 13, 2020
WYOMING						

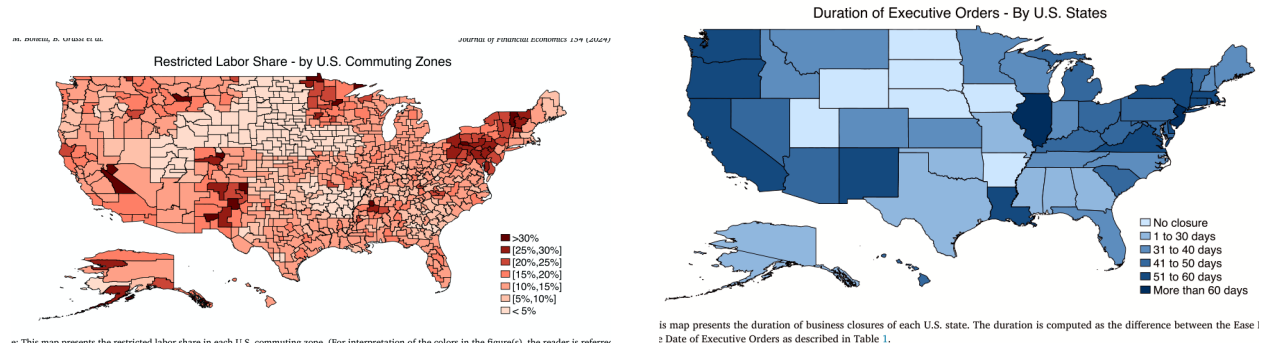
Table 1: Timing of initial Executive Orders restricting business activities. The table documents the state-by-state timing used to construct the closure treatment.

5.2 Figure 1: Restricted Labor Share by U.S. Commuting Zones; Figure 2: Duration of Executive Orders by U.S. States

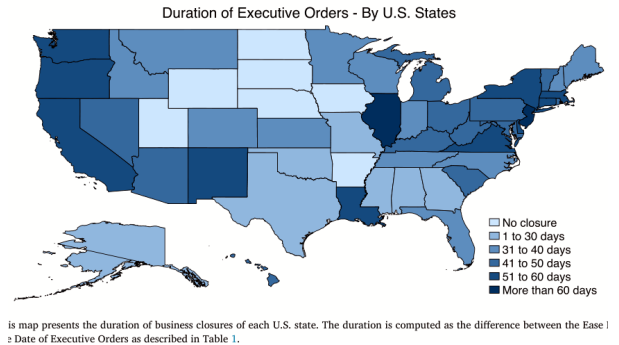
Read: Figure 1 maps restricted labor shares across CZs; Figure 2 maps closure duration by state.

Economic message: The figures show treatment variation from both local industry composition and state policy duration.

Detailed interpretation: Figure 1 shows within-state CZ heterogeneity, while Figure 2 shows state-level differences in closure length. The paper's treatment combines both sources.



(a) Figure 1: Restricted Labor Share by U.S. Commuting Zones.



(b) Figure 2: Duration of Executive Orders by U.S. States.

5.3 Table 2: Summary statistics

Read: The table summarizes 10,702 firm-quarter observations, 739 CZs, 27,994 CZ-week observations, and 4,800 CZ-month observations.

Economic message: Table 2 establishes the scale and variation of the firm, labor, and health samples.

Detailed interpretation: The restricted-labor distribution shows that a 10 percentage point exposure is economically meaningful and within the observed variation.

Table 2

Summary statistics. Panel A of this table presents summary statistics for our firm sample at the quarterly frequency over the sample period 2019Q1-2020Q3. All variables are winsorized at the first and ninety-ninth percentiles of their distributions. Panels B, C, and D present summary statistics for our CZ sample. Panel B presents the variables for each of the 739 CZs. Panel C (respectively Panel D) presents outcome variables and the restricted labor share at the weekly frequency (respectively monthly frequency) across the 739 CZs over the sample period 2020Jan-2020October. All continuous variables are winsorized at the first and ninety-ninth percentiles of their distributions.

	count	mean	sd	p1	p50	p99
Panel A: Firm level						
Assets/Assets ₂₀₁₈	10,702	1.190	0.411	0.519	1.084	2.890
Sales/Sales ₂₀₁₈	10,702	1.072	0.440	0.084	1.019	3.305
Income/Assets ₂₀₁₈	10,702	0.015	0.060	-0.262	0.025	0.131
Restricted Labor $\times I_{ShutDown}$	10,702	0.029	0.120	0.000	0.000	0.767
Panel B: CZ level						
Restricted Labor Share	739	0.092	0.083	0.000	0.102	0.330
Pop Density	739	0.093	0.160	0.000	0.039	0.995
Share Urban Pop	739	0.506	0.254	0.000	0.529	0.967
Migration 2019	739	-0.001	0.009	-0.024	-0.001	0.021
Employment Ratio	739	0.458	0.062	0.299	0.458	0.621
Work-From-Home Share	739	0.269	0.045	0.111	0.268	0.382
Contact-Intensive Share	739	0.591	0.030	0.525	0.591	0.683
Kids Share	739	0.132	0.017	0.090	0.132	0.198
Share Less High School Diploma 2018	739	0.086	0.035	0.034	0.080	0.187
Share 65+ Years Old	739	0.187	0.042	0.107	0.183	0.292
Median HH Income 2018 ('000s)	739	52.377	10.674	32.135	51.197	86.604
ICU beds/Pop	739	0.172	0.135	0.000	0.166	0.619
Ln(1+NbHospitals)	739	1.840	0.875	0.000	1.792	4.466
Donald Trump Vote Share (2016)	739	0.611	0.143	0.249	0.624	0.862
Migration 2020 (March-September)/Pop 2019	739	-0.002	0.017	-0.045	-0.002	0.042
Panel C: CZ-Week level						
New Covid-19 Infections per 10,000	27,994	7.387	12.774	0.000	2.435	60.104
New Covid-19 Deaths per 10,000	27,994	0.133	0.295	0.000	0.000	1.351
New Covid-19 Hospitalizations per 10,000	27,994	3.337	5.617	0.000	1.534	27.966
Log(Employment)	27,994	10.899	1.470	7.684	10.772	14.495
Log(Wages)	27,994	15.872	1.669	11.610	15.873	19.815
Log(Hours)	27,994	14.084	1.529	10.588	13.995	17.772
Restricted Labor $\times I_{ShutDown}$	27,994	0.044	0.072	0.000	0.008	0.275
$I_{ShutDown}$	27,994	0.043	0.204	0.000	0.000	1.000
Stay At Home Share	27,994	0.155	0.360	0.000	0.000	1.000
Mask Share	27,994	0.246	0.401	0.000	0.000	1.000
Panel D: CZ-Month level						
Total Deaths per 10,000	4,800	8.126	2.775	2.079	8.024	15.080
Years of Life Lost per 10,000	4,800	126.300	44.972	30.141	124.563	239.098
Quality-adjusted Years of Life Lost per 10,000	4,800	96.620	34.187	23.178	95.144	182.338

Table 2: Summary statistics.

5.4 Table 3: Restricted labor and firms' outcomes; Figure 3: Dynamics of firm sales, assets, and profits

Read: A 10 percentage point increase in restricted labor reduces sales by about 2.3%, assets by about 1.2%, and profits/assets by about 0.16 percentage points.

Economic message: Closures impose real firm-level losses.

Detailed interpretation: The event study supports parallel trends and shows that firm outcomes fall after restrictions rather than before.

Table 3

Restricted labor and firms' outcomes. This table presents estimates from panel regressions of individual firm sales (Panel A), assets (Panel B) and profits (Panel C) on the restricted labor share of the firm when business closures are active (2020-Q1 to 2020-Q3). In Panel A, sales are scaled by the firm's sales in 2018 in the same quarter of the year. In Panel B and C, assets and profits are scaled by the firm's assets in 2018 in the same quarter of the year. All regressions include firm fixed effects. We present results with quarter fixed effects (column 1), sector-quarter fixed effects (column 2), state-quarter fixed effects (column 3) and both sector-quarter and state-quarter fixed effects (column 4). In column 5, we control for the share of the firm employment that is subject to county-level mask mandates and stay-at-home orders. Standard errors presented in parentheses are clustered both at the firms' main industry and state levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Sales/Sales ₂₀₁₈					
Restricted Labor $\times I_{\text{Shut Down}}$	-0.306*** (0.057)	-0.213** (0.091)	-0.319*** (0.058)	-0.225** (0.087)	-0.227** (0.088)
Quarter FE	Yes	No	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Sector $\times$ Quarter FE	No	Yes	No	Yes	Yes
State $\times$ Quarter FE	No	No	Yes	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	Yes
Obs.	10,702	10,702	10,702	10,702	10,702
R ²	0.597	0.651	0.615	0.660	0.660
Panel B: Assets/Assets ₂₀₁₈					
Restricted Labor $\times I_{\text{Shut Down}}$	-0.087 (0.057)	-0.091 (0.055)	-0.126*** (0.045)	-0.119** (0.053)	-0.119** (0.055)
Quarter FE	Yes	No	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Sector $\times$ Quarter FE	No	Yes	No	Yes	Yes
State $\times$ Quarter FE	No	No	Yes	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	Yes
Obs.	10,702	10,702	10,702	10,702	10,702
R ²	0.726	0.759	0.736	0.765	0.765
Panel C: Profits/Assets ₂₀₁₈					
Restricted Labor $\times I_{\text{Shut Down}}$	-0.016*** (0.006)	-0.015** (0.007)	-0.017*** (0.005)	-0.015** (0.007)	-0.016** (0.007)
Quarter FE	Yes	No	No	No	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Sector $\times$ Quarter FE	No	Yes	No	Yes	Yes
State $\times$ Quarter FE	No	No	Yes	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	Yes
Obs.	10,702	10,702	10,702	10,702	10,702
R ²	0.806	0.838	0.813	0.842	0.842

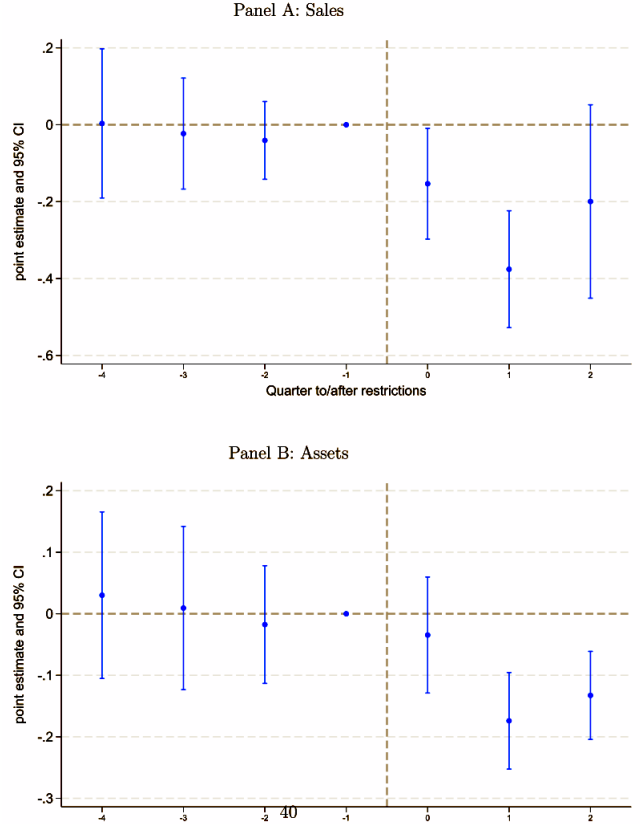
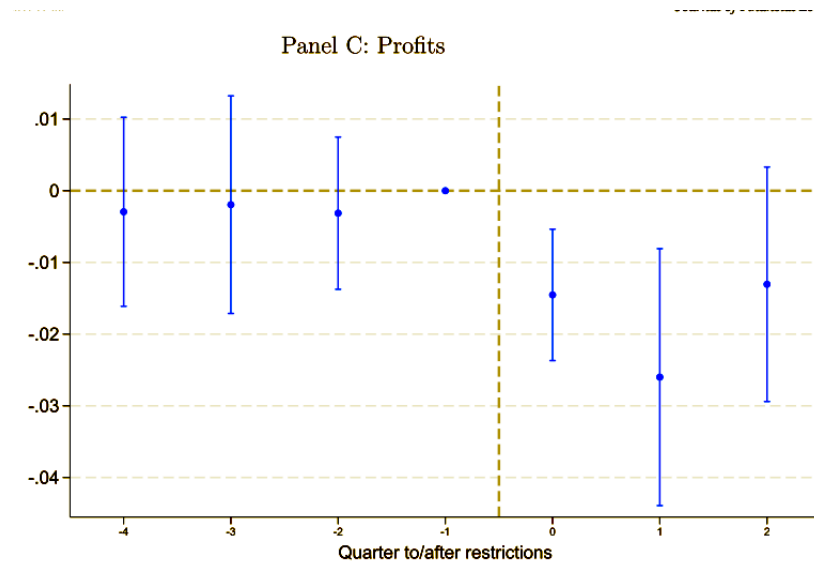
(a) Table 3: Restricted labor and firms' outcomes.**(b)** Figure 3: Dynamics of sales and assets.

Figure 3 continued: Dynamics of profits.

5.5 Table 4: Restricted labor, wages, employment and hours; Figure 4: Dynamics of wages, employment, and hours

Read: A 10 percentage point increase in restricted labor lowers wages by about 9.6%, employment by about 5.7%, and hours by about 8.2%.

Economic message: Workers are hit, but aggregate losses are smaller than firm profit losses, implying partial insurance by firms.

Detailed interpretation: The stability of coefficients after stay-at-home and mask controls indicates that the labor effects are not simply due to other social-distancing policies.

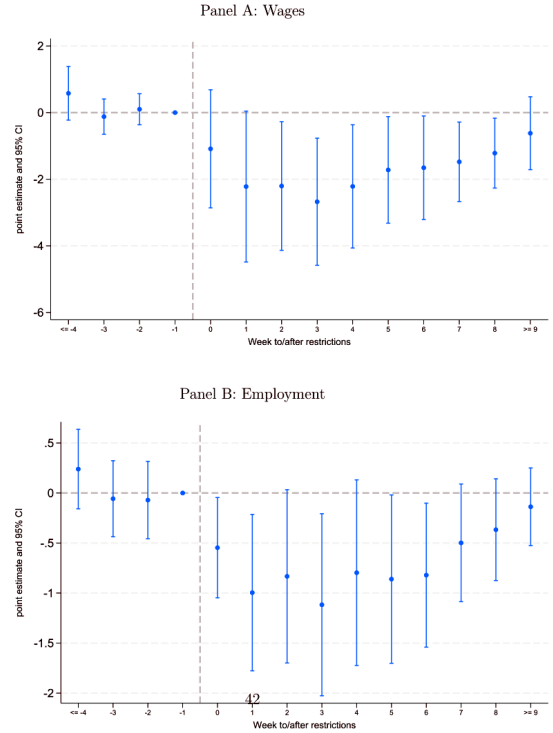
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Table 4
Restricted labor, wages, employment and hours. This table presents estimates from panel regressions of log(wages) (Panel A), log(employment) (Panel B), and log(hours) (Panel C) on the restricted labor share interacted with a dummy that equals one when business closures are active. Restricted labor in Panel A is computed using weights based on the total payroll of a given industry in each CZ whereas it is computed using employment weights in Panel B and C. All regressions include CZ fixed effects, state $\times$ week fixed effects, and in the second to sixth columns additional controls interacted with a dummy that equals one if business closures have been active. In the second to sixth columns, we include the urban share, population density, initial infection rates, and the share of work-from-home occupations in a given CZ. In the third to sixth columns, we include the share of contact-intensive occupations, as well as the kids share. In the fourth to sixth columns, we include net migrations in 2019, the share of the population with less than a high school diploma, the share of the population above 65 years old, and the logarithm of median household income. In the fifth and sixth column, we add the log of the number of hospitals plus one, the number of ICU beds per inhabitants and Donald Trump Vote Share in 2016. In the sixth column, we include the share of the commuting zone population that is subject to local mask and stay-at-home orders. Standard errors presented in parentheses are clustered both at the state level and at the week level. Regressions are population-weighted. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Log(Wages)						
Restricted Labor $\times I_{ShutDown}$	-0.763** (0.441)	-0.970*** (0.385)	-0.982*** (0.392)	-0.906*** (0.408)	-0.937*** (0.398)	-0.956*** (0.398)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{ShutDown}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{ShutDown}$	No	No	Yes	Yes	Yes	Yes
Census Controls $\times I_{ShutDown}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{ShutDown}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.992	0.992	0.992	0.992	0.992	0.992
Panel B: Log(Employment)						
Restricted Labor $\times I_{ShutDown}$	-0.538** (0.269)	-0.566*** (0.246)	-0.549*** (0.246)	-0.559*** (0.254)	-0.541*** (0.246)	-0.560*** (0.246)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{ShutDown}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{ShutDown}$	No	No	Yes	Yes	Yes	Yes
Census Controls $\times I_{ShutDown}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{ShutDown}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.995	0.996	0.996	0.996	0.996	0.996
Panel C: Log(Hours)						
Restricted Labor $\times I_{ShutDown}$	-0.782*** (0.295)	-0.797*** (0.273)	-0.787*** (0.272)	-0.822*** (0.288)	-0.798*** (0.278)	-0.822*** (0.275)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{ShutDown}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{ShutDown}$	No	No	Yes	Yes	Yes	Yes
Census Controls $\times I_{ShutDown}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{ShutDown}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.995	0.995	0.995	0.995	0.995	0.995

(a) Table 4: Restricted labor, wages, employment and hours.



(b) Figure 4: Dynamics of wages and employment.

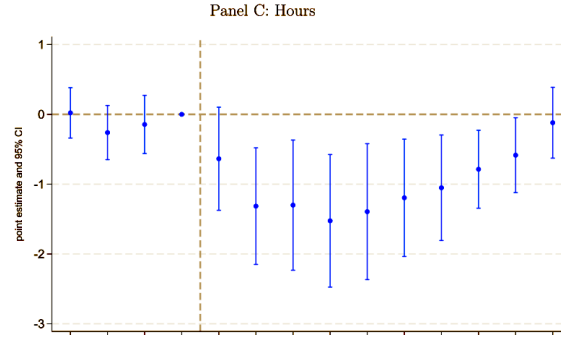


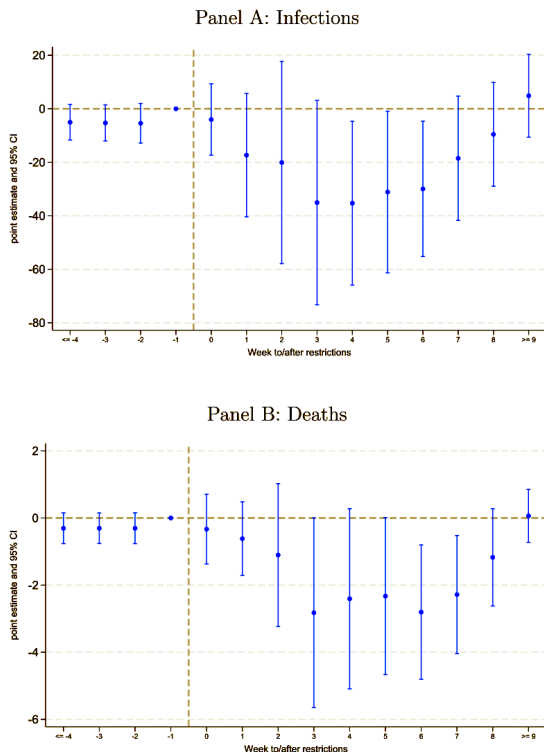
Figure 4 continued: Dynamics of hours.

5.6 Table 5: Restricted labor, Covid-19 infections and deaths; Figure 5: Dynamics of Covid-19 infections and deaths

Read: Restricted labor reduces new Covid-19 infections and deaths per 10,000 inhabitants, with effects occurring after implementation.

Economic message: Closures have measurable health benefits.

Detailed interpretation: The lagged outcome structure reflects the timing of disease transmission: infections respond first, deaths later.



(a) Figure 5: Dynamics of Covid-19 infections and deaths.

Table 5
Restricted labor, Covid-19 infections and deaths. This table presents estimates from panel regressions of new Covid-19 infections (Panel A) and new Covid-19 deaths (Panel B), both per 10,000 inhabitants, on the restricted labor share interacted with a dummy that equals one when business closures are active. All regressions include CZ fixed effects, state $\times$ week fixed effects, lagged infection rates, and in the second to sixth columns additional controls interacted with a dummy that equals one if business closures have been active. In the second to sixth columns, we include the urban share, population density, initial infection rates, and the share of work-from-home occupations in a given CZ. In the third to sixth columns, we include the share of contact-intensive occupations, as well as the kids share. In the fourth to sixth columns, we include net migrations in 2019, the share of the population with less than a high school diploma, the share of the population above 65 years old, and the logarithm of median household income. In the fifth and sixth column, we add the log of the number of hospitals plus one, the number of ICU beds per inhabitants and Donald Trump Vote Share in 2016. In the sixth column, we include the share of the commuting zone population that is subject to local mask and stay-at-home orders. Standard errors presented in parentheses are clustered both at the state level and at the week level. Regressions are population-weighted. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Infections						
New Covid-19 Infections per 10,000 (T+1)						
Restricted Labor $\times I_{ShareHours}$	-7.974** (3.726)	-8.218** (3.826)	-8.347** (3.935)	-7.526* (3.907)	-7.467** (3.883)	-7.142** (4.029)
New Covid-19 Infections per 10,000	0.722*** (0.046)	0.722*** (0.046)	0.721*** (0.046)	0.714*** (0.046)	0.714*** (0.046)	0.714*** (0.046)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{ShareHours}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{ShareHours}$	No	No	Yes	Yes	Yes	Yes
Census Controls $\times I_{ShareHours}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{ShareHours}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.901	0.901	0.901	0.901	0.901	0.901
Panel B: Deaths						
New Covid-19 Deaths per 10,000 (T+2)						
Restricted Labor $\times I_{ShareHours}$	-1.887*** (0.634)	-1.936*** (0.613)	-1.999*** (0.617)	-1.928*** (0.610)	-1.957*** (0.613)	-2.000*** (0.610)
New Covid-19 Infections per 10,000	0.020*** (0.005)	0.020*** (0.005)	0.020*** (0.005)	0.019*** (0.005)	0.019*** (0.005)	0.019*** (0.005)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{ShareHours}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{ShareHours}$	No	No	Yes	Yes	Yes	Yes
Census Controls $\times I_{ShareHours}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{ShareHours}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.786	0.787	0.788	0.790	0.790	0.790

might in turn lead us to overestimate the effect of business on death rates. To address this concern, we use data collected by Bloom (2021) who submitted a freedom of information request to the United States Postal Service's National Change of Address (NCOA) dataset and obtain zip code-month level inflow and outflow data at the CZ-level for the period ranging from September 2020. We then adjust the 2019 CZ-level population to scale Covid-19 deaths for net migrations (inflows minus outflows). In Online Appendix Table A.4, we present the results of this adjustment and find that the results are virtually identical to our baseline. Hence, we conclude that net migrations do not appear to be driving or biasing our findings.

ent literature in econometrics has raised concerns about the validity of negative weights in two-period difference-in-differences designs when treatment timing is staggered and there exists hetero-

geneity in the treatment effect (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). The average of these state-specific estimates, which do not use common weights, shows a similar pattern to our main estimates, with negative post-closure estimates on deaths (see Online Appendix Table A.5).

Finally, our baseline estimates are potentially biased estimates of true effects of mandated business closures on death rates in the absence of business closures. Formally, the validity of the identification strategy relies on the assumption that CZs are the relevant geographical units for the health outcomes in one CZ are not affected by business closures in neighboring areas, the so-called "no-interference" (Rubin) or "stable unit treatment value" (Angrist et al., 1996) assumption. This assumption is already taken into account treatment effect estimates across areas within the same CZ, which is arguably where the strongest given that workers are more likely to commute within than outside CZs. Still, we now turn to e-

(b) Table 5: Restricted labor, Covid-19 infections and deaths.

5.7 Table 6: Restricted labor, total deaths, and years of life lost; Table 7: Restricted labor and Covid-19 hospitalizations

Read: The mortality evidence extends beyond reported Covid deaths to total deaths, life-years, QALYs, and hospitalizations.

Economic message: The health effects are robust to broader mortality measures.

Detailed interpretation: Total deaths and years of life lost reduce concerns that the result is only about Covid-death coding.

Table 6

Restricted labor, total deaths, and years of life lost. This table presents estimates from panel regressions of new total deaths (Panel A), years of life lost (Panel B), and quality-adjusted years of life lost (Panel C), all per 10,000 inhabitants, on the restricted labor share interacted with a dummy that equals one when business closures are active. All regressions include CZ fixed effects, state $\times$ month fixed effects, lagged infection rates, and in the second to sixth columns additional controls interacted with a dummy that equals one if business closures have been active. In the second to sixth columns, we include the urban share, population density, initial infection rates, and the share of work-from-home occupations in a given CZ. In the third to sixth columns, we include the share of contact-intensive occupations, as well as the kids share. In the fourth to sixth columns, we include net migrations in 2019, the share of the population with less than a high school diploma, the share of the population above 65 years old, and the logarithm of median household income. In the fifth and sixth column, we add the log of the number of hospitals plus one, the number of ICU beds per inhabitants and Donald Trump Vote Share in 2016. In the sixth column, we include the share of the commuting zone population that is subject to local mask and stay-at-home orders. Standard errors presented in parentheses are clustered at the state level and at the month level. Regressions are population-weighted. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Total Deaths (Monthly)						
New Deaths per 10,000 (M+1)						
Restricted Labor $\times I_{\text{share}}^{\text{share}}$	-12.112** (4.639)	-12.015** (4.701)	-12.363** (4.687)	-12.388** (4.843)	-12.515** (4.962)	-12.747** (4.959)
New Covid-19 Infections per 10,000	0.043** (0.018)	0.043** (0.018)	0.042* (0.019)	0.043 (0.024)	0.043 (0.023)	0.043* (0.023)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{\text{share}}^{\text{share}}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{\text{share}}^{\text{share}}$	No	No	Yes	Yes	Yes	Yes
Consensus Controls $\times I_{\text{share}}^{\text{share}}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{\text{share}}^{\text{share}}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	4,800	4,800	4,800	4,800	4,800	4,800
R ²	0.897	0.897	0.897	0.897	0.898	0.898
Panel B: Years of Life (Monthly)						
New Years of Life Lost per 10,000 (M+1)						
Restricted Labor $\times I_{\text{share}}^{\text{share}}$	-160.999** (57.863)	-159.132** (59.176)	-165.200** (58.953)	-163.692** (61.740)	-165.893** (63.508)	-168.439** (63.880)
New Covid-19 Infections per 10,000	0.627** (0.233)	0.627** (0.235)	0.625** (0.251)	0.630 (0.339)	0.630 (0.310)	0.630* (0.304)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{\text{share}}^{\text{share}}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{\text{share}}^{\text{share}}$	No	No	Yes	Yes	Yes	Yes
Consensus Controls $\times I_{\text{share}}^{\text{share}}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{\text{share}}^{\text{share}}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	4,800	4,800	4,800	4,800	4,800	4,800
R ²	0.899	0.899	0.900	0.900	0.900	0.900
Panel C: Quality-adjusted Years of Life (Monthly)						
New Quality-adjusted Years of Life Lost per 10,000 (M+1)						
Restricted Labor $\times I_{\text{share}}^{\text{share}}$	-123.978** (45.466)	-122.435** (46.488)	-127.093** (46.388)	-125.853** (48.422)	-127.493** (49.791)	-129.473** (50.077)
New Covid-19 Infections per 10,000 (T)	0.481** (0.181)	0.481** (0.182)	0.479** (0.197)	0.484 (0.264)	0.484* (0.243)	0.484* (0.238)
Initial Infection, Share Urban, Pop Density, work-at-home $\times I_{\text{share}}^{\text{share}}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{\text{share}}^{\text{share}}$	No	No	Yes	Yes	Yes	Yes
Consensus Controls $\times I_{\text{share}}^{\text{share}}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{\text{share}}^{\text{share}}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.748	0.755	0.755	0.756	0.756	0.756

(a) Table 6: Restricted labor, total deaths, and years of life lost.

Table 7

Restricted labor and Covid-19 hospitalizations. This table presents estimates from panel regressions of new Covid-19 hospitalizations per 10,000 inhabitants on the restricted labor share interacted with a dummy that equals one when business closures are active. All regressions include CZ fixed effects, state $\times$ week fixed effects, lagged infection rates, and in the second to sixth columns additional controls interacted with a dummy that equals one if business closures have been active. In the second to sixth columns, we include the urban share, population density, initial infection rates, and the share of work-from-home occupations in a given CZ. In the third to sixth columns, we include the share of contact-intensive occupations, as well as the kids share. In the fourth to sixth columns, we include net migrations in 2019, the share of the population with less than a high school diploma, the share of the population above 65 years old, and the logarithm of median household income. In the fifth and sixth column, we add the log of the number of hospitals plus one, the number of ICU beds per inhabitants and Donald Trump Vote Share in 2016. In the sixth column, we include the share of the commuting zone population that is subject to local mask and stay-at-home orders. Standard errors presented in parentheses are clustered both at the state level and at the week level. Regressions are population-weighted. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

New Covid-19 Hospitalizations per 10,000 (T+1)						
Restricted Labor $\times I_{\text{share}}^{\text{share}}$	-8.348** (3.505)	-9.203** (3.317)	-9.604** (3.236)	-9.594** (3.294)	-9.656** (3.378)	-9.469** (3.434)
New Covid-19 Infections per 10,000	0.028*** (0.007)	0.029*** (0.007)	0.029*** (0.007)	0.028*** (0.007)	0.028*** (0.007)	0.028*** (0.006)
Initial Infection, Share Urban, Pop Density, Work-from-home $\times I_{\text{share}}^{\text{share}}$	No	Yes	Yes	Yes	Yes	Yes
Contact-Intensive Share, Kids Share $\times I_{\text{share}}^{\text{share}}$	No	No	Yes	Yes	Yes	Yes
Consensus Controls $\times I_{\text{share}}^{\text{share}}$	No	No	No	Yes	Yes	Yes
Hospitals, ICU Beds, Trump Share $\times I_{\text{share}}^{\text{share}}$	No	No	No	No	Yes	Yes
Stay at Home and Mask Share	No	No	No	No	No	Yes
CZ FE	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	27,994	27,994	27,994	27,994	27,994	27,994
R ²	0.748	0.755	0.755	0.756	0.756	0.756

(b) Table 7: Restricted labor and Covid-19 hospitalizations.

5.8 Table 8: Restricted labor and death outcomes: Low versus high-contact CZ

Read: Health benefits are larger in high-contact CZs.

Economic message: Targeted closures in high-contact areas may save lives at lower economic cost.

Detailed interpretation: This table moves from average effects to policy design: where workplace contact intensity is high, the health benefit of closures is much larger.

8

restricted labor and death outcomes: Low versus high-contact CZ. This table presents estimates from panel regressions of new Covid-19 deaths per 100,000 inhabitants at the CZ-week level, and estimates from panel regressions of new total deaths, years of life lost, and quality-adjusted years of life lost at the CZ-month level, on the restricted labor share interacted with a dummy that equals one when business closures are active. All regressions include CZ fixed effects, state $\times$ week fixed effects (respectively state $\times$ month fixed effects in the second to fourth column), lagged infection rates, lagged infection rates interacted with a dummy that equals one if business closures have been active, namely the urban share, population density, infection rates, the share of work-from-home occupations, the share of contact-intensive occupations, the kids share, net migrations in 2019, the share of the population with less than a high school diploma, the share of the population above 65 years old, the logarithm of median household income, the number of hospitals plus one, the number of ICU beds per inhabitants, Donald Trump Vote Share in 2016, and the share of the commuting population that is subject to local mask and stay-at-home orders. Standard errors presented in parentheses are clustered both at the state level and the CZ level in the first column, at the state level and at the month level in the second to fourth columns. Regressions are population-weighted. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

	CZ-week level	CZ-month level		
	Covid-19 Deaths (T+2) per 10,000	Total Deaths (M+1) per 10,000	Years of Life Lost (M+1) per 10,000	Quality-adjusted Years of Life Lost (M+1) per 10,000
restricted Labor $\times I_{StateClosures}$	-1.263*** (0.375)	-8.553* (4.172)	-105.082 (56.662)	-77.973 (42.733)
restricted Labor $\times I_{StateClosures} \times$ High contact CZ	-0.658*** (0.125)	-3.387** (1.353)	-51.173* (26.648)	-41.859* (20.720)
new Covid-19 Infections per 10,000	0.019*** (0.005)	0.043* (0.022)	0.627* (0.291)	0.481* (0.228)
lagged Infection, Share Urban, Population, Work-from-home $\times I_{StateClosures}$, contact-intensive Share, Kids Share	Yes	Yes	Yes	Yes
lagged Infection, Share Urban, Population, Work-from-home $\times I_{StateClosures}$, contact-intensive Share, Kids Share	Yes	Yes	Yes	Yes
lagged Infection, Share Urban, Population, Work-from-home $\times I_{StateClosures}$, contact-intensive Share, Kids Share	Yes	Yes	Yes	Yes
lagged Infection, Share Urban, Population, Work-from-home $\times I_{StateClosures}$, contact-intensive Share, Kids Share	Yes	Yes	Yes	Yes

Table 8: Restricted labor and death outcomes: Low versus high-contact CZ.

5.9 Table 9: Aggregate effects

Read: Aggregate losses are around \$475–\$580 billion depending on assumptions; SIRD estimates imply large health benefits.

Economic message: Table 9 is the paper’s cost-benefit synthesis.

Detailed interpretation: The table should be read as a range of model-dependent estimates, not a single welfare number. The robust insight is that both economic losses and health benefits are large.

Table 9

Aggregate effects. This table presents estimates of the aggregate effect of business closures on firms' profits, labor income, Covid-19 deaths, total deaths, the number of quality-adjusted years of life saved, the number of hospitalizations saved, and the cost of mental health. Several estimates are presented based on different specifications or assumptions. 95% confidence intervals are presented in squared brackets, as well as bounds to alternative assumptions about possible violations of parallel trends in the pre-treatment period using the approach proposed by Rambachan and Roth (2023).

Panel A: Profits and labor income (losses in bn \$)						Rambachan and Roth (2023)
	Baseline		Dynamic		$\bar{M} = 0.5$	
Firm profits						
Baseline	359	[51, 667]	392	[112, 673]	[0, 785]	
Assuming 10% profit-shifting	323	[46, 600]	352	[101, 606]	[0, 707]	
Labor income						
Homebase Wages	173	[31, 314]	186	[33, 339]	[-89, 588]	
Homebase Emp	186	[31, 342]	147	[24, 271]	[-119, 449]	
BLS Emp	152	[112, 191]				
Total losses						
Profits+labor income (Homebase Wages)	532	[194, 871]	579	[259, 899]		
Profits+labor income (Homebase Emp)	545	[201, 891]	540	[234, 847]		
Profits+labor income (BLS Emp)	511	[200, 821]				
Assuming 10% profit-shifting						
Profits+labor income (Homebase Wages)	496	[185, 807]	539	[244, 835]		
Profits+labor income (Homebase Emp)	510	[192, 828]	501	[220, 782]		
Profits+labor income (BLS Emp)	475	[195, 755]				
Share of profits in total losses (%)						
Profits/(Profits+labor income (Homebase Wages))	67%	[24%, 92%]	68%	[35%, 92%]		
Profits/(Profits+labor income (Homebase Emp))	66%	[22%, 92%]	73%	[40%, 95%]		
Profits/(Profits+labor income (BLS Emp))	70%	[25%, 83%]				
Assuming 10% profit-shifting						
Profits/(Profits+labor income (Homebase Wages))	65%	[22%, 92%]	65%	[33%, 92%]		
Profits/(Profits+labor income (Homebase Emp))	63%	[20%, 92%]	71%	[38%, 94%]		
Profits/(Profits+labor income (BLS Emp))	68%	[23%, 81%]				

Panel B: Health benefits (# in thousand)						Rambachan and Roth (2023)	
	SIRD		Linear		Dynamic	$\bar{M} = 0.5$	
Deaths							
Covid-19 Deaths	502	[202, 801]	184	[49, 319]	82	[13, 151]	[-18, 205]
Total Deaths	800	[190, 1409]	303	[45, 560]			
QALY	8118	[1964, 14271]	3172	[504, 5841]			
Hospitalizations	2375	[687, 4063]	765	[268, 1263]			
Mental Health (-)			5953	[4202, 7703]			

Panel C: Health benefits (monetary value in bn \$)						Rambachan and Roth (2023)	
	SIRD		Linear		Dynamic	$\bar{M} = 0.5$	
Covid-19 Deaths							
VSL = \$1.3 million	652	[262, 1042]	240	[64, 414]	107	[17, 197]	[-23, 266]
VSL = \$3 million	1505	[605, 2404]	552	[147, 957]	246	[39, 453]	[-54, 615]
VSL = \$6 million	3009	[1210, 4809]	1104	[294, 1914]	492	[78, 906]	[-108, 1230]
Total Deaths							
VSL = \$1.3 million	1039	[247, 1831]	394	[58, 728]			
VSL = \$3 million	2398	[569, 4226]	909	[135, 1680]			
VSL = \$6 million	4795	[1139, 8451]	1818	[270, 3360]			
QALY							
Value _{QALY} = \$50,000	406	[98, 714]	158	[25, 292]			
Value _{QALY} = \$100,000	812	[196, 1427]	317	[50, 584]			
Value _{QALY} = \$200,000	1624	[393, 2854]	634	[100, 1168]			
Hospitalizations							
Cost _{Hosp} = \$19,900	47	[14, 81]	15	[5, 25]			
Cost _{Hosp} = \$21,700	52	[15, 88]	17	[6, 27]			
Cost _{Hosp} = \$24,500	58	[17, 99]	18	[6, 30]			
Mental Health (-)							
Cost _{MDD} = \$1,900			11	[15, 8]			
Cost _{MDD} = \$6,800			40	[52, 28]			
Cost _{MDD} = \$24,000			142	[185, 101]			

Table 9: Aggregate effects

6 Short summary

- The paper studies U.S. state-mandated business closures during the first wave of Covid-19.
- The treatment is the share of labor restricted by sector-state closure rules after adjusting for work-from-home ability.
- Closures reduce firm sales, assets, and profits.
- Closures reduce wages, employment, and hours.
- Closures reduce infections, deaths, total mortality, life-years lost, QALYs lost, and hospitalizations.
- Firms bear about two thirds of aggregate economic losses, suggesting partial insurance of workers.
- Health benefits are concentrated in high-contact CZs.

- The cost-benefit conclusion depends on the SIRD versus linear health model and on value-of-life assumptions.

7 Conclusion

7.1 Core conclusion

Business closures had causal effects on both sides of the pandemic trade-off. They imposed substantial losses on firms and workers, but they also significantly reduced mortality and hospitalizations.

7.2 Incidence and insurance

The aggregate incidence results suggest that firms absorbed a large share of the cost through lower profits. Workers were hurt through lower wages, employment, and hours, but firms bore roughly two thirds of the aggregate economic burden.

7.3 Health interpretation

The health estimates show effects on reported Covid deaths, total deaths, years of life lost, QALYs, and hospitalizations. This strengthens the claim that closures had real mortality benefits.

7.4 Targeting implication

The high-contact CZ results imply that targeted closures could improve the cost-benefit trade-off by focusing restrictions where workplace contact creates the largest health externalities.

7.5 Limitations

The analysis is ex post and first-wave specific. It cannot fully capture long-run learning, vaccination, compliance differences, or all behavioral responses. The welfare comparison also depends on modeling assumptions and value-of-life estimates.

7.6 Takeaway

The paper provides a disciplined empirical framework for evaluating pandemic business closures. Its central lesson is that closure policies were both costly and effective, so the policy problem is not whether closures have effects, but how to target them efficiently.